

Karthik Seelam Sreenivas

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Education

Purdue University

Master of Science in Robotics Engineering

West Lafayette IN

May 2027

Visvesvaraya Technological University

Bachelor of Engineering in Mechanical Engineering

India

August 2022

Skills

- **Vehicle Systems:** ECU Development & Calibration, Automotive Electronics, CAN, Sensor Integration, HIL/SIL Testing, Vehicle Validation, Controls & Calibration, ADAS, Simulink
- **Programming:** Python, C/C++, MATLAB, Bash, Git, Linux
- **Controls & Robotics:** PID, MPC, Sensor Fusion, State Estimation, ROS 2, Gazebo
- **Engineering:** Vehicle Controls, Data Analysis, System Testing, Requirements Validation, CAD/CAM, Project management, R&D

Work Experience

IDEAS Lab

West Lafayette, Indiana

Graduate Student Researcher

Aug 2026 – Present

- Develop a VLA-based perception and goal-selection system for autonomous AGV and drone docking in unknown environments.
- Fuse camera, LiDAR, and IMU observations for probabilistic state estimation and implement uncertainty-aware trajectory tracking in ROS 2.

Canadian National Railway

Chicago, Illinois

Locomotive Onboard Technology INTERN

May 2026 – Aug 2026

- Built a Python and AI chatbot automation to diagnose Distributed Power (DP) locomotive system faults from operational logs.
- Developed a real-time GNSS health-monitoring system for locomotive GPS/INS hardware, detecting positioning degradation and antenna faults using an FMEA-driven fault-detection pipeline with IMU-based dead-reckoning fallback.

Purdue: Digital Twin Lab

West Lafayette, Indiana

Graduate Student Researcher

Jan 2026 – May 2026

- Developed a ROS 2 trajectory recording and replay system for self-driving vehicles using Autoware Universe and AWSIM, capturing human driving trajectories and autonomously reproducing them using MPC lateral and PID longitudinal control.

Bosch

Bangalore, India

Vehicle Calibration Engineer and Project Manager (R&D)

Aug 2022 – Aug 2025

- Led ECU software development, calibration, and controls activities across 6 vehicle and engine R&D programs spanning EU Stage V, Japan, and US Tier 4F emissions platforms, coordinating with software, OBD, air-system, and OEM/client management teams.
- Developed a pattern-based offline calibration methodology for ECU maps using historical vehicle data, enabling optimized map calibration prior to vehicle implementation and reducing calibration time and fuel consumption during vehicle testing.
- Developed and calibrated vehicle feature control algorithms for torque structure, ACC, cold-start strategies, PID/controller tuning, emission optimization, and RADAR Gen 5 safety features.
- Performed sensor integration and controller tuning across SIL, HIL, and vehicle testing; planned and conducted 3 fleet validations under extreme weather conditions.

Academic Projects

Terrain Planning and Control for Rovers Using CNN

Oct 2025 – Nov 2025

- Built a CNN-based semantic segmentation system (computer vision driven dataset) to classify diverse terrain types and infer dynamic properties (slope, friction, traversability) directly from onboard camera, RADAR, LiDAR and IMU data.
- Designed a physics-informed dynamic cost map that evaluates real-time terrain difficulty, enabling computation of both fastest and energy-optimal routes across unstructured environments.
- Implemented A* and RRT* planners in Gazebo to generate safe, efficient paths for rescue and extraterrestrial exploration rovers under varying terrain conditions. Global path is generated within 10 seconds, and a local planner generates the obstacle avoided path instantly.

Extracurriculars/Leadership

Technical & Suspension Team Head - Shwethashwa Racing | Formula SAE India

Jan 2020 – Dec 2022

- I led a team of 30 students for my undergrad college's automotive club as Technical and Suspension Team Head.
- Built a formula student car powered by a 390-cc engine supported with an indigenously developed direct actuating shocks and pull-rod suspensions for competing in SAE-SUPRA, Buddh International Circuit, Delhi, India.
- We won the best livery design of formula car for Formula Bharat Simulation Racing Competition in 2021.

Achievements

Starkhacks (AMD)

April 2026

- Won 2nd place at the world's largest hardware hackathon (Starkhacks) - made a physical AI inspired robot arm buddy that helps doing physical tasks on your desk and is also a companion who would not let the user feel lonely.